Lecture 3. Basic Parametric Distributions and Models

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1 Outline

- Basic parametric distributions for survival data: exponential, Weibull, extreme value, log-normal, log-logistic, gamma.
- Popular regression models for survival data: accelerated failure time (log-linear), multiplicative (proportional hazards), additive hazards.

2 Exponential Distribution

Definition 3.1: Exponential(λ)

$f(x) = \lambda e^{-\lambda x}$, $S(x) = e^{-\lambda x}$, $h(x) = \lambda$ (constant), $H(x) = \lambda x$.

- The conditional probability of failure at any time t, given no event before t, does not depend on t (memoryless).
- $H(x) = -\ln S(x) = \lambda x$ gives an **empirical check**: plot an estimate of $H(x)$ against x; a straight line through the origin supports the exponential model.
- Mathematically simple but too restrictive in both health and industrial applications.
- Special case of both the Weibull ($\alpha = 1$) and the gamma ($\beta = 1$).

3 Weibull Distribution

Definition 3.2: Weibull(λ, α), $\lambda, \alpha > 0$

$$h(x) = \lambda\alpha x^{\alpha-1}, \qquad H(x) = \lambda x^{\alpha}, \qquad S(x) = \exp(-\lambda x^{\alpha}).$$

- Mathematically simple, yet fairly flexible: increasing hazard ($\alpha > 1$), decreasing ($\alpha < 1$), constant ($\alpha = 1$).
- **Empirical check**: $\ln H(x) = \ln\lambda + \alpha \ln x$, so plot $\ln[\hat{H}(x)]$ against $\ln x$ and look for linearity (slope α, intercept $\ln\lambda$).

4 Weibull and the Extreme Value Distribution

Motivation. The exponential model has constant hazard, which is rarely realistic. The Weibull distribution adds a shape parameter so the hazard can increase or decrease, and on the log scale it becomes a location–scale model, the basis of accelerated failure time regression.

Definition 3.3: Standard extreme value distribution

$$f_W(w) = \exp\{w - e^w\}, \qquad S_W(w) = \exp\{-e^w\}, \qquad -\infty < w < \infty.$$

It is a special case of the Gompertz family.

Theorem 3.4: Log-linear representation of the Weibull

If $Y \equiv \ln X = \mu + \sigma W$ with W standard extreme value, then $X \sim \text{Weibull}(\lambda, \alpha)$ with

$$\alpha = \frac{1}{\sigma}, \qquad \lambda = e^{-\mu/\sigma} \quad \Big(\text{equivalently } \mu = \frac{-\ln\lambda}{\alpha},\ \sigma = \frac{1}{\alpha}\Big).$$

Proof [Supplement]. First, $S_W(w) = \int_w^\infty e^{v-e^v}\,dv$; substituting $u = e^v$ gives $\int_{e^w}^\infty e^{-u}\,du = \exp(-e^w)$. Then

$$S_X(x) = P(\ln X > \ln x) = P\Big(W > \frac{\ln x - \mu}{\sigma}\Big) = \exp\Big\{-e^{(\ln x-\mu)/\sigma}\Big\} = \exp\Big\{-e^{-\mu/\sigma}x^{1/\sigma}\Big\},$$

which is $\exp(-\lambda x^{\alpha})$ with $\alpha = 1/\sigma$, $\lambda = e^{-\mu/\sigma}$. ■

Hence, if the error in a log-linear model is standard extreme value, X is Weibull.

5 Log-normal Distribution

Definition 3.5

If $Y = \ln X \sim N(\mu, \sigma^2)$ then $X \sim$ log-normal(μ, σ^2): $S(x) = 1 - \Phi\Big(\frac{\ln x - \mu}{\sigma}\Big)$.

- Close tie to the normal distribution.
- Can fit unimodal (hump-shaped) hazards: the hazard starts at 0, rises to a maximum and then decreases.

- Criticized as a lifetime distribution because the hazard *always decreases at the tail* (to 0).
- May still fit cases where large values of x are not of interest.

6 Log-logistic Distribution

Motivation. Some hazards rise and then fall, for example after surgery. The log-logistic distribution allows this and has closed-form survival and hazard functions.

Definition 3.6: Log-logistic

If $Y = \ln X = \mu + \sigma W$ with W standard logistic, $\mathrm{LG}(0,1)$:

$$f_W(w) = \frac{e^w}{(1+e^w)^2}, \qquad S_W(w) = \frac{1}{1+e^w},$$

then $Y \sim \mathrm{LG}(\mu, \sigma^2)$ with $f_Y(y) = \dfrac{e^{(y-\mu)/\sigma}}{\sigma\left(1+e^{(y-\mu)/\sigma}\right)^2}$ and $X \sim$ log-logistic.

Derivation: survival function of X

$$S_X(x) = S_W\Big(\frac{\ln x - \mu}{\sigma}\Big) = \frac{1}{1+\exp\left(\frac{\ln x-\mu}{\sigma}\right)} = \frac{1}{1+e^{-\mu/\sigma}x^{1/\sigma}} = \frac{1}{1+\lambda x^\alpha},$$

with $\alpha = 1/\sigma$ and $\lambda = e^{-\mu/\sigma}$.

Theorem 3.7: Log-logistic hazard

$$h_X(x) = \frac{\lambda\alpha x^{\alpha-1}}{1+\lambda x^\alpha}.$$

It is monotone decreasing when $\alpha \le 1$. When $\alpha > 1$ it increases to a maximum at $x^* = \{(\alpha-1)/\lambda\}^{1/\alpha}$ and then decreases to 0 as $x \to \infty$.

Proof [Supplement]. $H(x) = -\ln S(x) = \ln(1+\lambda x^\alpha)$, so $h(x) = H'(x) = \lambda\alpha x^{\alpha-1}/(1+\lambda x^\alpha)$. Differentiating,

$$h'(x) = \lambda\alpha\,\frac{(\alpha-1)x^{\alpha-2}(1+\lambda x^\alpha) - x^{\alpha-1}\lambda\alpha x^{\alpha-1}}{(1+\lambda x^\alpha)^2} = \lambda\alpha\,\frac{x^{\alpha-2}\left[(\alpha-1) - \lambda x^\alpha\right]}{(1+\lambda x^\alpha)^2}.$$

If $\alpha \le 1$ the bracket is negative for all $x > 0$. If $\alpha > 1$ the bracket is positive for $\lambda x^\alpha < \alpha - 1$ and negative afterwards, giving the maximum at x^*; and $h(x) \sim \alpha/x \to 0$. ∎

The hazard resembles the log-normal one but has a closed form.

7 Gamma Distribution

Definition 3.8: Gamma(λ, β), $\lambda, \beta > 0$

$$f(x) = \frac{\lambda^\beta x^{\beta-1}e^{-\lambda x}}{\Gamma(\beta)}, \qquad S(x) = \int_x^\infty \frac{\lambda^\beta s^{\beta-1}e^{-\lambda s}}{\Gamma(\beta)}\,\mathrm{d}s, \qquad h(x) = \frac{x^{\beta-1}e^{-\lambda x}}{\int_x^\infty s^{\beta-1}e^{-\lambda s}\,\mathrm{d}s}.$$

No closed form for S or h in general.

- $\beta = 1$: exponential(λ). $\beta \to \infty$: approximately normal. $\lambda = 1/2$, integer β: $\chi^2_{2\beta}$.
- $\beta > 1$: h is monotone increasing with $h(0) = 0$ and $h(x) \to \lambda$ as $x \to \infty$.
- $\beta < 1$: h is monotone decreasing with $h(0) = \infty$ and $h(x) \to \lambda$ as $x \to \infty$.
- Not as mathematically tractable as the others; not widely used (the Weibull is more popular).

Derivation: why the gamma hazard tends to λ monotonically

Substitute $s = x + u$ in the reciprocal hazard:

$$\frac{1}{h(x)} = \int_x^\infty \left(\frac{s}{x}\right)^{\beta-1} e^{-\lambda(s-x)}\,\mathrm{d}s = \int_0^\infty \left(1 + \frac{u}{x}\right)^{\beta-1} e^{-\lambda u}\,\mathrm{d}u.$$

As $x \to \infty$, $(1 + u/x)^{\beta-1} \to 1$, so $1/h(x) \to \int_0^\infty e^{-\lambda u}\,\mathrm{d}u = 1/\lambda$. For $\beta > 1$, $(1 + u/x)^{\beta-1}$ decreases in x, so $1/h$ decreases and h increases; for $\beta < 1$ the reverse holds.

8 Regression Models for Survival Data

To deal with heterogeneity in the population, the quantities summarizing survival must be adjusted for covariates (explanatory variables). For example, in non-randomized breast cancer studies the two groups may be imbalanced in age, tumour size, number of positive nodes, or hormonal status.

8.1 Approach 1: Log-linear / accelerated failure time (AFT) model

Definition 3.9: Log-linear model

$$Y \equiv \ln X = \mu + \boldsymbol{\gamma}'\mathbf{z} + \sigma W,$$

where $W \sim F$ (a known distribution) and $\mu, \sigma, \boldsymbol{\gamma}$ are unknown.

Theorem 3.10: Log-linear ⇒ AFT

Let $X_0 = \exp(\mu + \sigma W)$ with survival S_0. Then

$$S(x \mid \mathbf{z}) = S_0(x\, e^{-\boldsymbol{\gamma}'\mathbf{z}}), \qquad h(x \mid \mathbf{z}) = \eta\, h_0(\eta x), \quad \eta = e^{-\boldsymbol{\gamma}'\mathbf{z}}.$$

Proof [Supplement].

$$\begin{aligned} P(X > x \mid \mathbf{z}) &= P(Y > \ln x \mid \mathbf{z}) = P(\mu + \sigma W > \ln x - \boldsymbol{\gamma}'\mathbf{z}) \\ &= P\big(\exp(\mu + \sigma W) > x \exp(-\boldsymbol{\gamma}'\mathbf{z})\big) \\ &= S_0(x e^{-\boldsymbol{\gamma}'\mathbf{z}}). \end{aligned}$$

For the hazard, $S(x \mid \mathbf{z}) = S_0(\eta x)$ gives $f(x \mid \mathbf{z}) = \eta f_0(\eta x)$ (chain rule), so

$$h(x \mid \mathbf{z}) = \eta f_0(\eta x)/S_0(\eta x) = \eta h_0(\eta x).$$

■

- The covariate rescales time: time is **accelerated** if $\eta = e^{-\boldsymbol{\gamma}'\mathbf{z}} > 1$ and **decelerated** if $\eta < 1$.
- The log-linear model induces the AFT model but with **opposite signs** for the regression coefficients: a positive γ lengthens survival ($\eta < 1$).

Derivation: interpretation of γ

For covariate values $\mathbf{z}_1, \mathbf{z}_2$, since $X = e^{\mu+\boldsymbol{\gamma}'\mathbf{z}} e^{\sigma W}$,

$$\frac{\mathrm{E}(X \mid \mathbf{z}_1)}{\mathrm{E}(X \mid \mathbf{z}_2)} = \frac{e^{\mu+\boldsymbol{\gamma}'\mathbf{z}_1}\mathrm{E}(e^{\sigma W})}{e^{\mu+\boldsymbol{\gamma}'\mathbf{z}_2}\mathrm{E}(e^{\sigma W})} = e^{\boldsymbol{\gamma}'(\mathbf{z}_1-\mathbf{z}_2)} \implies \ln \frac{\mathrm{E}(X \mid \mathbf{z}_1)}{\mathrm{E}(X \mid \mathbf{z}_2)} = \boldsymbol{\gamma}'(\mathbf{z}_1 - \mathbf{z}_2).$$

So γ is the log of the ratio of mean survival times for a unit increase in z. (The same holds for any quantile.)

Definition 3.11: Three choices of F

1. $W \sim N(0,1)$: $\ln X \sim N(\mu + \boldsymbol{\gamma}'\mathbf{z}, \sigma^2)$, i.e. X is **log-normal**.
2. W standard extreme value, $f_W(w) = \exp(w - e^w)$: $X \sim \text{Weibull}(\lambda, \alpha)$ with $\alpha = 1/\sigma$ and $\lambda = \exp\{-(\frac{\mu}{\sigma} + \frac{\boldsymbol{\gamma}'\mathbf{z}}{\sigma})\}$.
3. W standard logistic: the **log-logistic** regression model.

8.2 Approach 2: Multiplicative (proportional hazards) model

Definition 3.12: Multiplicative hazards model

$$h(x \mid \mathbf{z}) = h_0(x)\, C(\boldsymbol{\beta}'\mathbf{z}),$$

where the baseline hazard $h_0(x) \geq 0$ may be parametric or left unspecified. With $C(\boldsymbol{\beta}'\mathbf{z}) = e^{\boldsymbol{\beta}'\mathbf{z}}$ this is the **Cox proportional hazards model**.

- β is the **log hazard ratio**, constant over time: $h(x \mid \mathbf{z}_1)/h(x \mid \mathbf{z}_2) = e^{\boldsymbol{\beta}'(\mathbf{z}_1-\mathbf{z}_2)}$.
- Integrating, $H(x \mid \mathbf{z}) = e^{\boldsymbol{\beta}'\mathbf{z}} H_0(x)$ and $S(x \mid \mathbf{z}) = \exp\{-e^{\boldsymbol{\beta}'\mathbf{z}} H_0(x)\} = \{S_0(x)\}^{e^{\boldsymbol{\beta}'\mathbf{z}}}$.
- **Graphical check** of PH: $\ln[-\ln S(x \mid \mathbf{z})] = \boldsymbol{\beta}'\mathbf{z} + \ln[-\ln S_0(x)]$, so log-cumulative-hazard curves for different $\mathbf{z}$ should be *parallel* (vertical shift $\boldsymbol{\beta}'\mathbf{z}$).

Example 3.13: Weibull baseline under PH

If $h_0(x) = \alpha\lambda x^{\alpha-1}$ then $h(x \mid \mathbf{z}) = \alpha\lambda x^{\alpha-1}\exp(\boldsymbol{\beta}'\mathbf{z})$ and

$$S(x \mid \mathbf{z}) = \exp\big[-\lambda \exp(\boldsymbol{\beta}'\mathbf{z}) x^{\alpha}\big], \quad \text{i.e.} \quad X \mid \mathbf{z} \sim \text{Weibull}(\lambda e^{\boldsymbol{\beta}'\mathbf{z}}, \alpha).$$

Theorem 3.14: Weibull is closed under both AFT and PH

The Weibull is the only continuous distribution that is closed under both the AFT and PH models.

Derivation: matching the two parametrizations

Under AFT with Weibull baseline, $h(x \mid \mathbf{z}) = \eta h_0(\eta x) = \eta\,\alpha\lambda(\eta x)^{\alpha-1} = \eta^\alpha \cdot \alpha\lambda x^{\alpha-1}$, a PH model with hazard ratio $\eta^\alpha = e^{-\alpha\boldsymbol{\gamma}'\mathbf{z}}$. Hence

$$\boldsymbol{\beta} = -\alpha\boldsymbol{\gamma} = -\boldsymbol{\gamma}/\sigma.$$

8.3 Approach 3: Additive hazards model

Definition 3.15: Additive hazards model

$$h(x \mid \mathbf{z}) = h_0(x) + \sum_{j=1}^{p} z_j(x)\beta_j(x).$$

- Allows covariate effects to vary over time.
- The regression functions may be positive or negative, so they must be constrained to keep $h(x \mid \mathbf{z}) \geq 0$.

9 Selected exercises

Motivation. Three textbook exercises (K&M Section 2.8): a family whose hazard can be bathtub shaped, the property that characterizes log-logistic regression, and a computation with a guarantee time.

Exercise 3.1 (K&M 2.4: exponential power distribution)

Let $S(x) = \exp\{1 - \exp[(\lambda x)^\alpha]\}$, $x \geq 0$, with $\lambda, \alpha > 0$.
(a) If $\alpha = 0.5$, show that the hazard is bathtub shaped and find where it changes from decreasing to increasing.
(b) If $\alpha = 2$, show that the hazard is monotone increasing.

Solution.

Step 1 (the hazard). $H(x) = -\log S(x) = \exp[(\lambda x)^\alpha] - 1$, which is 0 at $x = 0$, so

$$h(x) = H'(x) = \alpha\lambda^\alpha x^{\alpha-1}\exp[(\lambda x)^\alpha], \qquad x > 0.$$

Step 2 (its log-derivative). For $x > 0$,

$$\frac{\mathrm{d}}{\mathrm{d}x}\log h(x) = \frac{\alpha-1}{x} + \alpha\lambda^\alpha x^{\alpha-1} = \frac{(\alpha-1) + \alpha(\lambda x)^\alpha}{x}.$$

Since $h > 0$, h increases exactly where the numerator $N(x) = (\alpha-1) + \alpha(\lambda x)^\alpha$ is positive.

(a) For $\alpha = 0.5$, $N(x) = \frac{1}{2}\{(\lambda x)^{1/2} - 1\}$, negative for $x < 1/\lambda$ and positive for $x > 1/\lambda$. So h decreases on $(0, 1/\lambda]$ and increases on $[1/\lambda, \infty)$; moreover $h(x) \to \infty$ as $x \downarrow 0$ (factor $x^{-1/2}$) and as $x \to \infty$ (factor $e^{(\lambda x)^{1/2}}$). This is a bathtub shape with minimum at $x^* = 1/\lambda$. In general, for $\alpha < 1$ the turning point is $x^* = \lambda^{-1}\{(1-\alpha)/\alpha\}^{1/\alpha}$.

(b) For $\alpha = 2$, $N(x) = 1 + 2(\lambda x)^2 > 0$ for all x, so h is strictly increasing (more generally whenever $\alpha \geq 1$). ■

Exercise 3.2 (K&M 2.16: log-logistic regression is a proportional odds model)

Given a covariate Z, let $Y = \ln X = \mu + \gamma Z + \sigma W$ with $\sigma > 0$ and W standard logistic, $f_W(w) = e^w/(1+e^w)^2$.
(a) Find $S(x \mid Z)$.
(b) Find the odds of death before x, $[1 - S(x \mid Z)]/S(x \mid Z)$.
(c) Show that for two covariate values Z_1, Z_2 the ratio of these odds does not depend on x.

Solution.

(a) With $S_W(w) = 1/(1+e^w)$ (Definition of the log-logistic in this lecture),

$$S(x \mid Z) = P\Big(W > \frac{\ln x - \mu - \gamma Z}{\sigma}\Big) = \frac{1}{1 + \exp\{(\ln x - \mu - \gamma Z)/\sigma\}} = \frac{1}{1 + \lambda_Z x^\alpha},$$

where $\alpha = 1/\sigma$ and $\lambda_Z = \exp\{-(\mu + \gamma Z)/\sigma\}$.

(b) $1 - S(x \mid Z) = \lambda_Z x^\alpha/(1 + \lambda_Z x^\alpha)$, so the odds are $\lambda_Z x^\alpha$.

(c) The ratio is

$$\frac{\lambda_{Z_1} x^\alpha}{\lambda_{Z_2} x^\alpha} = \exp\Big\{-\frac{\gamma(Z_1 - Z_2)}{\sigma}\Big\},$$

free of x. Thus the log-logistic AFT model is also a *proportional odds* model, with log odds ratio $-\gamma(Z_1 - Z_2)/\sigma$, in the same way that the Weibull AFT model is a PH model with $\boldsymbol{\beta} = -\boldsymbol{\gamma}/\sigma$. ■

Exercise 3.3 (K&M 2.11: Weibull with a guarantee time)

For $\phi \geq 0$ let $S(x) = \exp[-\lambda(x-\phi)^\alpha]$ for $x > \phi$ and $S(x) = 1$ for $x \leq \phi$.
(a) Find the hazard and density.
(b) With $\alpha = 1$, $\lambda = 0.0075$, $\phi = 100$, find the mean and the median lifetime.

Solution.

(a) For $x \leq \phi$, $S \equiv 1$ gives $f(x) = h(x) = 0$. For $x > \phi$, $H(x) = \lambda(x-\phi)^\alpha$, so

$$h(x) = \alpha\lambda(x-\phi)^{\alpha-1}, \qquad f(x) = h(x)S(x) = \alpha\lambda(x-\phi)^{\alpha-1}\exp[-\lambda(x-\phi)^\alpha].$$

That is, $X - \phi \sim \text{Weibull}(\lambda, \alpha)$: no failure can occur before the guarantee time ϕ.

(b) With $\alpha = 1$, $X - 100 \sim \text{Exp}(0.0075)$. Hence

$$\text{E}(X) = 100 + \frac{1}{0.0075} = 233.3, \qquad x_{0.5} = 100 + \frac{\ln 2}{0.0075} = 100 + 92.4 = 192.4.$$

■